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Studierichting: Informatica
Oefeningenreeks 5E nr. 1

$$\begin{aligned} S &= \frac{1}{2} \int_0^{2\pi} \cancel{x^2} d\theta - \frac{1}{2} \int_0^\pi \cancel{\sin^2(\theta)} d\theta \frac{1 - \cos(2\theta)}{2} \\ &= \frac{1}{2} [\theta]_0^{2\pi} - \frac{1}{4} \left[\theta - \frac{1}{2} \sin(2\theta) \right]_0^\pi \\ &= \frac{1}{2} (2\pi - 0) - \frac{1}{4} ((\pi - 0) - (0 - 0)) \\ &= \pi - \frac{\pi}{4} = \frac{3\pi}{4} \end{aligned}$$

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References